

AB POWER ENGINEERING

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ENGINE OPERATION, REPAIR, MAINTENANCE, AND TROUBLE SHOOTING

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1. Space comprising of the clearance volume and the piston displacement volume of an engine. ☒ a. Combustion chamber ☐ b. Engine displacement volume ☐ c. Total volume – piston displacement volume ☐ d. None of the above 2. Engine capable of compressing air-fuel mixture in order to create explosion when ignited. ☒ a. Gasoline engine ☐ b. Diesel engine ☐ c. Flexi-Fuel Engine ☐ d. None of the above 3. Engine that raises the temperature of the air high enough during compression to create combustion of fuel. ☒ a. Gasoline engine ☐ b. Diesel engine ☐ c. Flexi Fuel Engine ☐ d. None of the above	4. Which of the following is an advantage of compression-ignition over spark-ignition engine? ☐ a. Higher reliability ☐ b. Uniform distribution of fuel in each cylinder ☐ c. Uniform torque characteristics ☒ d. All of the above ☐ e. None of the above 5. Which of the following is an advantage of two-stroke over 4-stroke engine? ☐ a. Power stroke is much higher for the same engine size and speed. ☐ b. Simple in construction. ☒ c. Higher mechanical efficiency. ☐ d. All of the above ☐ e. None of the above 6. Which of the following is a possible cause of engine's failure to start up? ☐ a. No gas ☐ b. Damage spark plugs ☒ c. Dirty/clogged air filter ☐ d. All of the above
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7. Which of the following is a possible cause of engine running poorly or stalling after starting? ☐ a. Governor needs adjustment. ☐ b. Engine accessories are jammed. ☒ c. All of the above ☐ d. None of the above 8. Excessive smoke while engine is running is due to ____. ☐ a. oil has reached exhaust manifold ☒ b. cracked cylinder head gasket ☐ c. All of the above ☐ d. None of the above 9. What factor causes overheating of engines? ☐ a. Blocked cooling fins ☐ b. Dirty or clogged air intake ☐ c. Faulty radiator and/or water pump ☒ d. All of the above ☐ e. None of the above	10. What factor causes knocking noise of engines. ☐ a. Carbon deposit inside the cylinder ☐ b. Engine has overheated ☒ c. Spark-plug damaged ☐ d. All of the above ☐ e. None of the above 11. Which of the following problems frequently experienced in diesel engines? ☐ a. Engine overheating ☐ b. Oil leakages ☐ c. Oil supply problem ☒ d. Using wrong fuel ☐ e. All of the above ☐ f. None of the above 12. What is the negative effect of overheating of engine. ☐ a. Damage to bearings and crankshaft ☐ b. Piston expanding and scraping against cylinder walls ☐ c. Cylinder head swelling, distorting , and braking ☒ d. All of the above
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13. Engine part that frequently encounters problems.

- a. Piston
- b. Gasket
- c. Cylinder head
- d. Bearings
- ☒ e. All of the above
- f. Three of the above

14. When diesel engine lacks power, the probable cause is ____.

- a. lack compression
- b. choked air supply
- c. heavy carbon deposit on the intake valve
- ☒ d. All of the above

15. Which of the following is a common cause of troubles among multiple-cylinder diesel engines?

- a. Abnormal exhaust
- b. Hard to start
- c. Lack of power
- d. Overheating

e. Defective instruments

- f. Air leakages
- g. Fuel leakages
- h. Malfunctioning of fuel injection equipment
- ☒ i. All of the above
- j. None of the above

16. When there is insufficient air during combustion of fuel, the color of the smoke coming out of the muffler is ____.

- a. blue
- ☒ b. black
- c. gray
- d. All of the above
- e. None of the above

17. When the lubricating oil is carried into the combustion chamber of the engine, the color of smoke will be ____.

- a. black
- ☒ b. blue
- c. gray
- d. All of the above

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18. When excessive amount of water is mixed with the fuel or there is leaky valve set rings, the color of the smoke is ____.

- a. black
- b. blue
- ☒ c. gray
- d. All of the above

19. What is the factor that causes heavy knocking in cylinders?

- a. Too early or too late injection
- b. Choked or enlarge nozzle holes
- c. Insufficient lubricating oil
- d. Too hot circulating water
- ☒ e. All of the above
- f. Three of the above

20. Which of these factors causes high-cooling water-outlet temperature?

- a. Low water pressure
- b. Water pump failure
- c. Engine too hot

d. Malfunctioning of thermostat valve.

- ☒ e. All of the above
- f. Three of the above

21. What factor causes the engine to run irregularly?

- a. Damage governor
- b. Water in fuel
- c. Leaky fuel pump valve
- d. Defective injection nozzle
- ☒ e. All of the above
- f. None of the above

22. What factor affects the frequency of changing engine oil.

- a. Type of engine (diesel or gasoline)
- b. Age of engine
- c. Operating condition of engine
- ☒ d. All of the above
- e. None of the above

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23. Frequency of changing engine oil for modern diesel farm tractors.

- a. 50 to 100 hours
- ☒ b. 100 to 250 hours
- c. 250 to 300 hours
- d. None of the above

24. Frequency of changing engine oil for older diesel farm tractors.

- ☒ a. 75 to 150 hours
- b. 150 to 250 hours
- c. 250 to 300 hours
- d. None of the above

25. Frequency of changing engine oil for gasoline-powered farm tractors.

- ☒ a. 50 to 100 hours
- b. 100 to 150 hours
- c. 150 to 200 hours
- d. None of the above

26. Frequency of changing engine oil for light-used tractors.

- a. 50 to 200 hours
- b. 200 to 350 hours
- ☒ c. At least once a year
- d. None of the above

27. Frequency of changing engine oil for single-piston gasoline engine used by farmers.

- a. Every months
- ☒ b. Every 6 months
- c. Every 12 months
- d. None of the above

28. Required oil change for initial breaking-in of engine to remove initial contaminants from manufacturing process.

- ☒ a. 50 hours
- b. 100 hours
- c. 150 hours
- d. None of the above

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29. Frequency of engine oil change in regular operation of single-piston gasoline engine.

- a. 50 to 100 hours
- b. 100 to 150 hours
- c. 150 to 200 hours
- d. None of the above

30. Engine subjected to harsh condition like dusty, hot climatic condition will required changing oil every ____ of operation.

- a. 25 hours
- b. 50 hours
- c. 75 hours
- d. None of the above

31. For routine use, single-piston diesel engines require changing of oil every ____.

- a. 100 to 200 hours
- b. 200 to 300 hours
- c. 300 to 400 hours
- d. Ne of the above

32. For heavy-duty use, single-piston diesel engines require changing of oil every ____.

- a. 50 to 100 hours
- b. 100 to 150 hours
- c. 150 to 200 hours
- d. None of the above

33. Which of the enumerated factors affects oil-change frequency in diesel engine?

- a. Type of use
- b. Oil type
- c. Manufacturer's recommendations
- d. Operating conditions
- e. Oil analysis
- f. Engine age and condition
- g. All of the above
- h. Four of the above

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34. Frequency of changing air filter in tractors.

- a. Every 500 hours of use
- b. Every 750 hours of use
- c. Every 1000 hours of use
- d. None of the above

35. Dusty or dirty environment may require ____ change of filter.

- a. more frequent
- b. less frequent
- c. seldom frequent
- d. None of the above

36. Which of the following factors affects frequency of air filter replacement?

- a. Operating environment
- b. Usage intensity
- c. Signs of clogged filter
- d. Manufacturer recommendation
- e. All of the above
- f. None of the above

37. Frequency of changing spark-plugs in small engines.

- a. Once a season
- b. Twice a season
- c. Thrice a season
- d. None of the above

38. Which among the enumerated factors affects spark-plug life in single-piston engines?

- a. Type of spark plug
- b. Operating condition
- c. Engine type
- d. Signs of wear
- e. All of the above
- f. None of the above

39. When changing spark plug for multiple-cylinder engine, it is recommended to change ____.

- a. them in one set
- b. only those who are damaged
- c. one at a time
- d. None of the above

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